

Get Set Go with Raspberry Pi Camera

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This project describes how to connect a web camera to the Raspberry Pi and make use of it. Raspberry Pi board has an onboard connector which can be used for connecting the camera directly. Connect the camera very carefully, as shown in Fig. 1, and take good grounding measures before touching it.

Installation of Raspberry Pi camera

First of all, open the Raspberry Pi's configuration file, Raspi config, with the command:

```
$ sudo raspi-config
```

Then enable the camera, as shown in Fig. 2.

If your firmware is old, update and upgrade it using the command:

```
$ sudo apt-get update
$ sudo apt-get upgrade
```

Now you should see the camera option. Enable it, as indicated in Fig. 2. After connecting the camera to the Raspberry Pi, see how best you can make use of it.

The command 'raspistill' is used to take still pictures. It comes with various parameters which you can choose for the command. Type raspistill in the terminal and you will see all the options, as shown in Table 1.

For example, if you want to capture an image, type the command given below (also refer Fig. 3):

```
$ raspistill -v -o image.jpg
```

Here '-v' indicates that the output is verbose. You will see a preview of the snapshot for five seconds and the snapshot will be saved as image.jpg in the home directory. You can



do several things with the image, such as defining its dimensions and quality. Check all the options and try them.

For video, you can use 'raspidv' command. For checking various options, type raspivid in the terminal. You will find all the options, as shown in Table II.

To capture a 10-second video with the Raspberry Pi camera, type the command (also refer Fig. 4):

```
$ raspivid -o video.h264 -t 10000
```

Here 'video' is the name of your video, '.h264' is the file extension and '10000' is the time in milliseconds.

Check all the functions; you can do several things, such as recording the video and defining its frame size. You can even stream the video captured by the camera. The simplest method that I found for this was with mplayer and netcat.

Download both mplayer and netcat on a Windows computer from the link:

<http://dl.dropboxusercontent.com/u/106074492/mplayer%20and%20netcat.zip>

Extract files and you will see two exe files (mplayer.exe and nc.exe) that can be executed through command.

Install netcat and mplayer on Raspberry



Fig. 1: Raspberry Pi camera

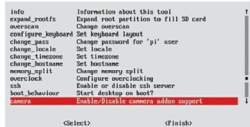


Fig. 2: Enabling the camera

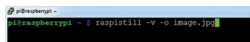


Fig. 3: Capturing an image



Fig. 4: Recording the video

TABLE I
Raspistill Options

Commands	Functions
Image Parameter Commands	
-w, --width	Set image width <size>
-h, --height	Set image height <size>
-q, --quality	Set jpeg quality <0 to 100>
-r, --raw	Add raw bayer data to jpeg metadata
-o, --output	Output filename <filename> (to write to stdout, use '-o -'). If not specified, no file is saved
-v, --verbose	Output verbose information during run
-t, --timeout	Time (in ms) before it takes picture and shuts down (if not specified, set to 5s)
-sh, --sharpness	Set image sharpness (-100 to 100)
-co, --contrast	Set image contrast (-100 to 100)
-br, --brightness	Set image brightness (0 to 100)
-sa, --saturation	Set image saturation (-100 to 100)
-rot, --rotation	Set image rotation (0-359)
-hf, --hflip	Set horizontal flip
-vf, --vflip	Set vertical flip
Preview Parameter Commands	
-p, --preview	Preview window settings <'x,y,w,h'>
-f, --fullscreen	Full-screen preview mode
-op, --opacity	Preview window opacity (0-255)
-n, --nopreview	Do not display a preview window